

Comparison: IoT vs AI vs Data Analytics

IoT جسمانی آلات (ہارڈویئر) کا نیٹ ورک ہے جو ماحول سے خام ڈیٹا جمع کرتا ہے۔

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 05 of 13

Comparison: IoT vs AI vs Data Analytics

LEARN IT

These three technologies are often used together, so students mix them up. The table below, from the source, compares them on four aspects.

Aspect	IoT	Artificial Intelligence (AI)	Data Analytics
Definition	Network of physical devices connected to the internet.	Technology enabling machines to think and decide.	Process of examining data to find patterns.
Main purpose	Data collection and automation.	Intelligent decision making and predictions.	Extracting meaningful insights.
Type of work	Works with physical devices (hardware).	Works with smart software systems.	Works with data (numbers, records).
Output	Raw data from the environment.	Smart decisions or actions.	Reports, graphs, insights.
Role in a smart system	Eyes and hands	Brain	Organiser / reporter
Pakistani example	Safe City cameras in Lahore/Karachi capturing traffic	Software recognising a car's number plate for an e-challan	Monthly report showing which roads have the most violations

WORKED EXAMPLE

One problem, three technologies — a K-Electric smart meter:

IoT: The smart electricity meter measures units used every 15 minutes and sends them over the mobile network. Output: raw readings.

Data Analytics: The company cleans and studies a month of readings and draws a graph showing that most electricity is used between 7 pm and 11 pm. Output: report/graph.

AI: Software learns each area's pattern, predicts tomorrow's demand, and decides how to plan load management. Output: smart decision.

REMEMBER IT

Ask "What is the output?" Raw readings → IoT. Charts and reports → Data Analytics. A decision or prediction → AI.

COMMON MISTAKE

Mistake: Writing that IoT "makes decisions". **Fix:** In the comparison table IoT's main purpose is *data collection and automation*; the *intelligent decision* belongs to AI.

اردو میں سمجھیں

IoT جسمانی آلات (ہارڈویئر) کا نیٹ ورک ہے جو ماحول سے خام ڈیٹا جمع کرتا ہے۔ AI ذہین سافٹ ویئر ہے جو فیصلے اور پیش گوئیاں کرتا ہے۔ ڈیٹا اینالٹکس ڈیٹا کا جائزہ لے کر رپورٹس اور گراف بناتا ہے۔ تینوں مل کر ایک اسمارٹ نظام بناتے ہیں۔

CHECK YOURSELF

1. Which of the three works mainly with hardware?
2. What is the output of Data Analytics?

Answers: 1. IoT. 2. Reports, graphs and insights.

Next lecture: 5.2 Uses of IoT in Different Areas